

Two of Us: Bridging Asymmetric Perspectives

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Figure 1: Two players with different perspectives collaborate to complete the stage.

ABSTRACT

Two of Us is an immersive collaborative work that investigates how self-understanding is shaped through cooperation with others under an asymmetric information structure. Through complementary roles that provide access to different viewpoints and information, participants' task performance depends on communication and mutual interpretation, transforming collaboration into a mirror for self-awareness and reflection. By leveraging presence and embodied agency in XR, the system is designed to externalize users' past psychological habits under real-time constraints and to simulate relational changes based on the interaction data that emerge in this process. This work presents XR as not merely a simulation medium, but as a reflective framework for analyzing how past, present, and future are interconnected and reconfigured in human collaboration.

Index Terms: virtual reality, asymmetric cooperation, social VR, large language model, AI-based analysis, self-awareness

1 PROJECT URL

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2 DESCRIPTION OF ARTWORK

2.1 Thematic Statement

Two of Us is an immersive collaborative game based on an asymmetric information structure that investigates how self-understanding is formed and transformed through cooperation and relationships with others. Jean-Paul Sartre conceptualized the self not as an isolated interior entity, but as something that can only be recognized through relationships with others, particularly through the gaze of others. Grounded in this perspective, *Two of Us* experimentally implements a virtual environment in which the presence of others becomes a prerequisite for self-recognition and collaboration. By presenting a situation in which two players must exchange clues and cooperate while accessing different viewpoints and information, the work transforms collaboration itself into a medium for self-reflection and self-awareness. Through this approach, the work reconsiders how immersive experiences can reconfigure human collaborative relationships. It also proposes an experimental framework for exploring how empathy and collaborative interactions evolve in virtual environments, as well as how future relationships may be simulated.

2.2 Artwork

Unlike films or novels, games generate divergent experiences through non-linear structures shaped by each player's perspective and disposition. In particular, virtual reality (VR) provides a high level of immersion, enabling users to perceive the in-game character not as a separate other but as themselves [3]. *Two of Us* combines these characteristics of VR with a collaborative structure, and is designed so that interactions between two players reveal a wider range of experiences, collaborative processes, and relational patterns.

The work explores how collaborative interaction expands into self-understanding and relational perception through a temporal framework that connects the past, present, and future. Its core

premise is that an immersive environment with concrete collaborative constraints, such as asymmetry and interdependence, can soften the performative social masks maintained in everyday contexts and elicit more immediate and embodied responses. The interactions generated under these conditions are structured not as isolated events but as a continuous flow, forming a structure in which individual dispositions and relational tendencies emerge over time.

In this temporal context, the past functions as a background of accumulated embodied habits and communication styles that influences present behavioral choices. The present refers to the collaborative performance itself within the VR environment, in which two players solve tasks in real time based on restricted information and differentiated roles. Data generated from these present actions and interactions extends into the layer of the future, serving as cues for reflectively considering potential relational trajectories and communication patterns. In particular, the future dimension of this work is framed not as a deterministic prediction, but as a simulation of relational possibilities grounded in the perspective of speculative design [1]. The collected interaction data are not used as indicators to evaluate the users or their relationship. Instead, they are used to simulate potential future relationships based on the players' communication styles and interaction patterns, functioning as a mediating mechanism through which the two users can reflect on their relationship and modes of communication.

The results are presented in the form of a report, and the two users reflect on their respective tendencies and communication styles by jointly reading and interpreting the outcomes. This shared interpretive process strengthens self-recognition mediated by the other, and empirically reveals how past habits, present interactions, and future possibilities can be connected and reconfigured. Through this, Two of Us suggests that eXtended Reality (XR) can function not merely as a simulation medium, but also as a tool for reflecting on relationships and self-understanding, while enabling users to think about and construct possible futures.

2.3 Prototype Implementation

The prototype for the exhibition demo was structured by dividing the experience into parts, allowing the session to be completed within approximately 10 minutes. Accordingly, the system was designed so that participants can immediately view the report following the short experience.

Based on the dialogue and interaction data generated during collaboration between the two players, analyses grounded in the Big Five personality factors are performed and presented in the form of a report that reflects the game and conversational context. The report is generated using a Large Language Model (LLM), and by adopting the visual form of a game results screen, the process of self-understanding is presented in a manner analogous to reviewing game outcomes.

The work is implemented as a two-player VR game developed in Unity 3D (6000.0.58f1), utilizing Photon Fusion¹ and the Vivox² Unity SDK to provide multiplayer functionality and voice chat. The demo exhibition is conducted using two Meta Quest 3 headsets and is optimized to maintain a frame rate of 90 fps or higher.

Game narrative follows a setting in which two characters, left as residual data in a temporary storage system due to a system error, attempt to escape. Information asymmetry is implemented through differentiated roles, as follows. The Explorer navigates a constrained space from a first-person perspective to solve puzzles, while the Prospector operates from a broader, more comprehensive viewpoint to control environmental elements and coordinate progression. These differences in perspective and control are designed to require discourse as a core tool for collaboration.

During play, runtime logging records each player's voice data, in-game actions, and mission events. Analysis pipeline uses CLOVA Speech Recognition³ to separate and transcribe the two participants' speech by speaker, after which utterances are segmented based on mission phases and PGC (Problem–Goal–Constraint) tags. Subsequently, analyses grounded in prior literature [2, 6, 5, 4] are performed using a LLM (OpenAI GPT-4o⁴). This process extracts Big Five–based key lexical events, LIWC (Linguistic Inquiry and Word Count) categories, utterance contexts, and conflict situations, and organizes temporal patterns and events related to conflict. Finally, the LLM synthesizes these analytical results to simulate the future relationship between the two players, and the outcomes are visualized and presented in the final report. Participant data are not stored or retained during the exhibition demo.

3 CONCLUSION

Two of Us is an immersive XR work that explores self-awareness and relational patterns as they are formed through asymmetric collaborative interaction. The work presents a report-based reflective experience that connects habits from the past, collaborative performance in the present, and possibilities for future relationships. Through this, it suggests that XR can function not merely as a simulation medium, but as a reflective medium that prompts reconsideration of relationships and temporality.

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